YAKEEN

NEET 2.0 2027

Physics | Lecture - 11

Basic Maths and Calculus



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Todays Goal

⇒ application of differentiation

⇒ Maxima / minima

⇒ Integration

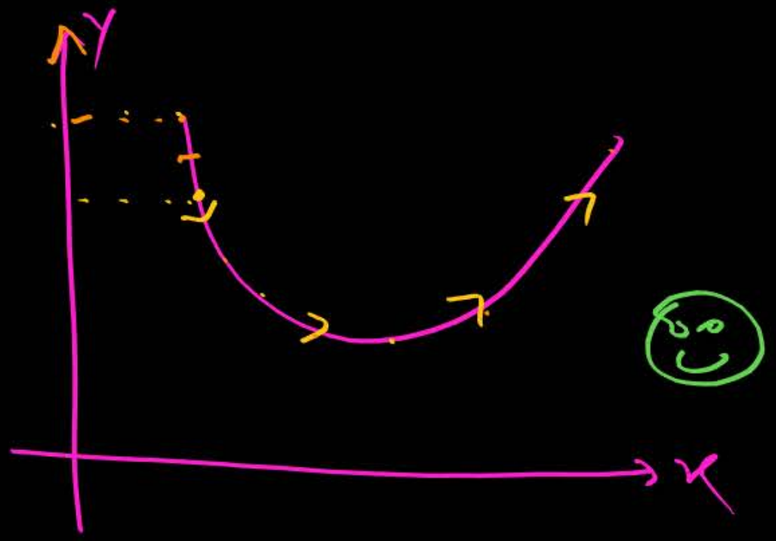
$\frac{dP}{dt}$ = Rate of change in P w.r.t time = force

$\frac{dy}{dx}$ = slope

$\frac{d|y|}{dx}$ = diff^n of magnitude of 'y' w.r.t 'x'
= Rate of change in magnitude of 'y'

$\left|\frac{dy}{dx}\right|$ = Magnitude of Rate of change in y w.r.t 'x'

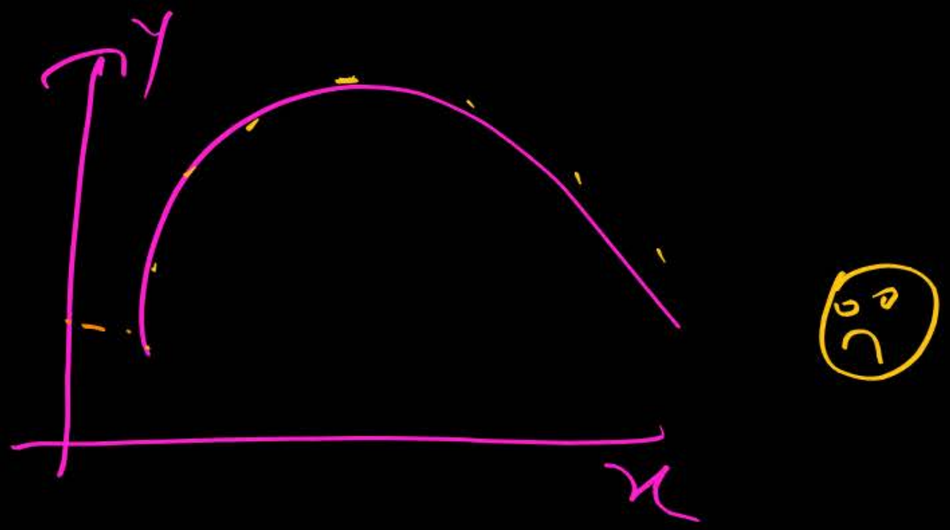
@ # $\frac{d^2y}{dx^2}$ = Rate of change in $\left(\frac{dy}{dx}\right)$ slope;



$y \rightarrow$ is 1st ↓ then y incⁿ ↑

$\frac{dx}{dx}$ = slope → always incre

$\left(\frac{d^2y}{dx^2}\right)$ → Change in Slope = tve



y is 1st ↑ then ↓

$\frac{dy}{dx}$ → slope continu ↓

$\frac{d^2y}{dx^2}$ → change in slope = -ve



@MRPHYSICSS

Copy

② gf Position of object $x = t^2 - 4t + 8$ then find velocity $\left(V = \frac{dx}{dt}\right)$ and draw V/t graph.

Soln

$$x = t^2 - 4t + 8$$

$$\frac{d(x)}{dt} = V = \frac{d(t^2 - 4t + 8)}{dt}$$

$$V = 2t - 4 \times 1 + 0$$

$$V = 2t - 4$$

initial velocity $V_{t=0} = 2 \times 0 - 4$

$= -4 \text{ m/s}$ ✓

$V_{t=2} = 2 \times 2 - 4 = 0$ ✓

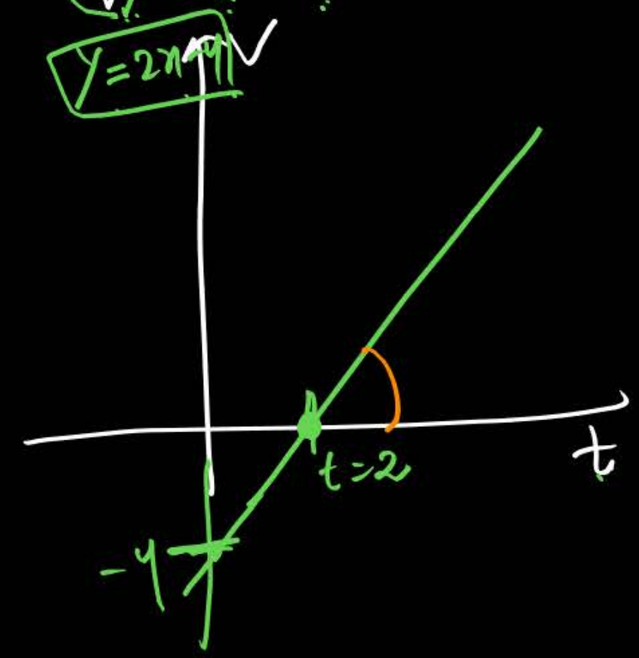
$V_{t=6} = 2 \times 6 - 4$

$= 12 - 4 = 8 \text{ m/s}$

Draw V/t graph:—

$V = 2t - 4$ $(Y = mx + c)$

$Y = 2x - 4$ ✓

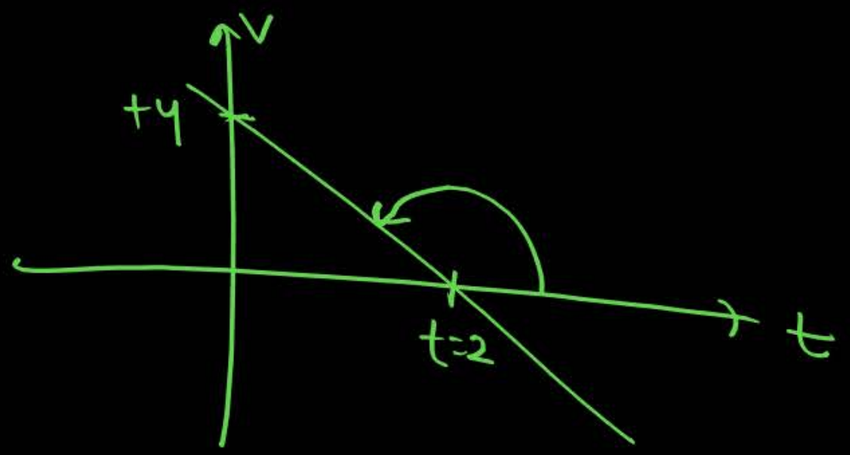


Q gf Position of object $x = -t^2 + 4t + 8$ then find velocity $V = \frac{dx}{dt}$ and draw V/t graph.

Soln

$$\frac{dx}{dt} = V = -[2t) + 4 \times 1 + 0$$

$$V = -2t + 4$$



Q) gf Velocity of object $V = \frac{t^3}{3} - 3t^2 + 8t$ then find acc^n $\left[a = \frac{dv}{dt}\right]$ and draw alt graph.

Soln

$$V = \frac{t^3}{3} - 3t^2 + 8t$$

$$a = \frac{dv}{dt} = \frac{3t^2}{3} - 3[2t] + 8 \times 1$$

$$a = t^2 - 6t + 8$$

$y = x^2 - 6x + 8$ (Parabolic)

$y_{intercept}$ $x = 0$

gf $a = 0$

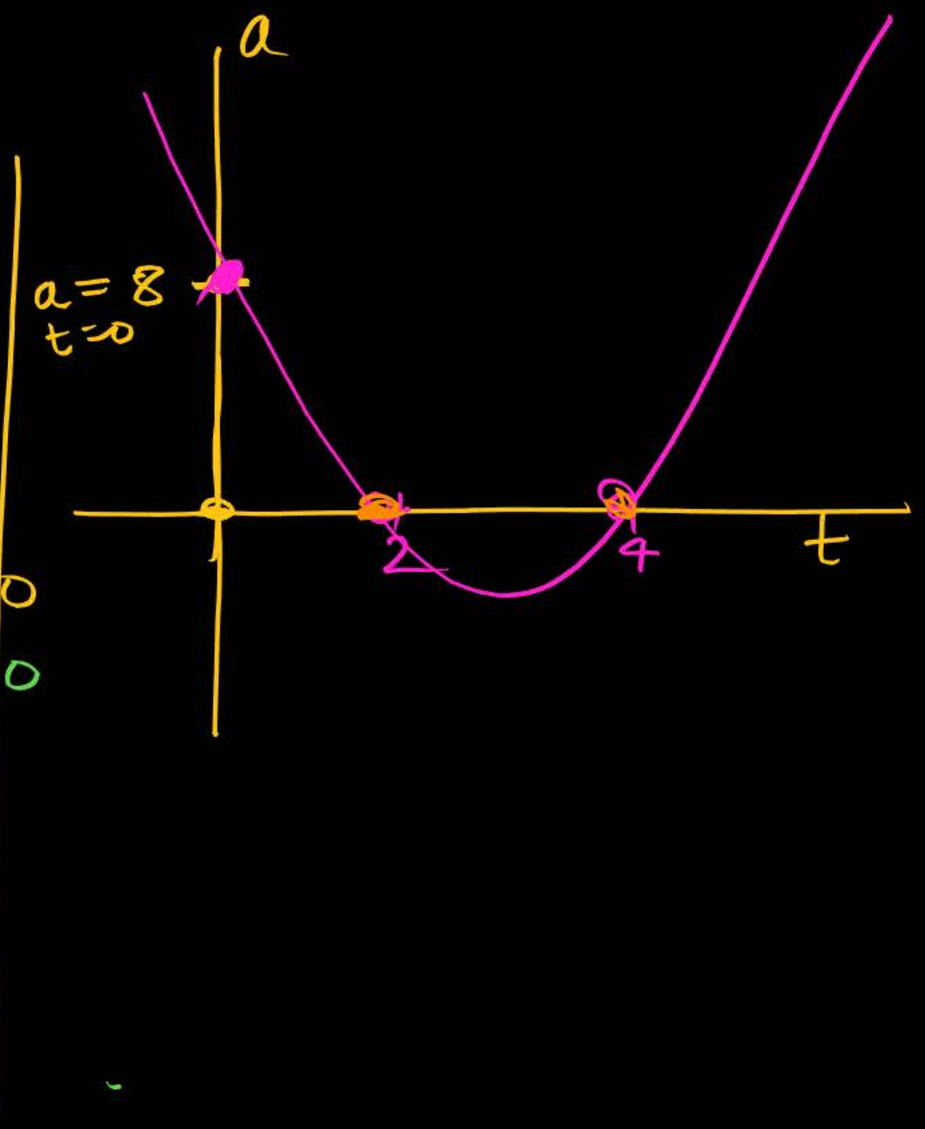
$0 = t^2 - 6t + 8$

$t^2 - 4t - 2t + 8 = 0$

$t(t-4) - 2(t-4) = 0$

$(t-4)(t-2) = 0$

$t_1 = 4 \quad t_2 = 2$



gf $y = x^2$ Find $\frac{d(y)}{dx}$ and $\frac{dy}{dt}$

diff^n w.r.t x

$$\frac{dy}{dx} = \frac{dx^2}{dx} = 2x$$

diff^n w.r.t 't'

$y = x^2$

$$\frac{d(y)}{dx} = \frac{dx^2}{dx} = 2x$$

$$\frac{dy}{dx} = 2x$$

$$\left[\frac{dy}{dt}\right] = 2x \times \left[\frac{dx}{dt}\right]$$

Ex1

$y = \sin(\theta)$

find $\frac{dy}{dt}$ =??

$\frac{dy}{d\theta} = \frac{d\sin\theta}{d\theta} = \cos\theta$

$\frac{dy}{d\theta} = \cos\theta$

$\boxed{\frac{dy}{dt} = \cos\theta \frac{d\theta}{dt}}$

(8) Volume of sphere $V = \frac{4}{3}\pi r^3$

r = radius

find $\frac{dV}{dt}$ =??

" $V = \frac{4}{3}\pi r^3$ "

diff^n w.r.t 't'

$\frac{dV}{dt} = \frac{4}{3}\pi \frac{dr^3}{dt}$

$\frac{dV}{dt} = \frac{4}{3}\pi \frac{dr^3}{dt} \times \frac{dr}{dr} = \frac{4}{3}\pi \frac{dr^3}{dr}\frac{dr}{dt}$

$= \frac{4}{3}\pi 3r^2 \left(\frac{dr}{dt}\right)$

$= 4\pi r^2 \frac{dr}{dt}$ ✓

$V = \frac{4}{3}\pi r^3$

$\frac{dV}{dt} = \frac{4}{3}\pi 3r^2 \left[\frac{dr}{dt}\right]$

$y = e^x$

$\frac{dy}{dx} = e^x$

$\frac{dy}{dt} = e^x \times \frac{dx}{dt}$ ✓

$$y = \sin(x^2+2x)$$

$$\frac{dy}{dt} = \cos(x^2+2x)\,(2x+2)\left(\frac{dx}{dt}\right)$$

$$A = 4\pi r^2$$

$$\frac{dA}{dt} = 4\pi\, 2r \left(\frac{dr}{dt}\right)$$ ✓

Slipe-12

Q/ If radius of circle is Increasing at rate $\frac{1}{\pi}$ m/sec then find rate of change in Area w.r.t time. when radius is 4m.

$$\frac{dr}{dt} = \frac{1}{\pi} \text{ m/sec}$$

$$\frac{dA}{dt} = ??$$

$r = 4m$

Area of cir

$$A = \pi r^2$$

$$\frac{dA}{dt} = \pi 2r \frac{dr}{dt}$$

$$\frac{dA}{dt} = \pi 2 \times 4 \times \frac{1}{\pi} = 8 \text{ m}^2/\text{sec}$$

The side of a square is Increasing at the rate of 0.2 cm/s. The rate of Increase of Perimetre with respect to time:-

(a) 0.4 cm/s

(b) 0.6 cm/s

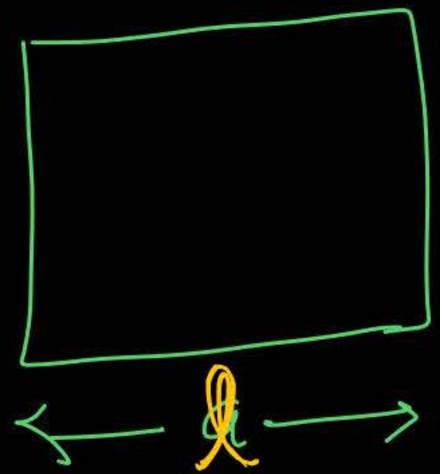
(c) 0.2 cm/s

(d) 0.8 cm/s ✓

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$\frac{dl}{dt} = 0.2$ cm/s.

$\frac{dP}{dt} = ??$



$P = 4l$

diff^n wrt t.

$\frac{dP}{dt} = 4\left(\frac{dl}{dt}\right) = 4 \times 0.2$

$= 0.8$ m/s.

274. If $V = \frac{4}{3}\pi R^3$; find rate of change in volume w.r.t. time

$\left(\frac{dV}{dt}\right) = ??$

$V = \frac{4}{3}\pi R^3$

$\frac{dV}{dt} = \frac{4}{3}\pi \cdot 3R^2 \left(\frac{dR}{dt}\right)$

223. Area of a circle is $A = \pi r^2$, where r is radius, then find the increasing rate of area with respect to radius:

(a) $2\pi r$ (b) πr^2 (c) 2π (d) πr

224. A balloon is being inflated such that its volume is increasing at a rate of 10 cubic centimeters per second. How fast is the radius of the balloon increasing when the radius is 5 centimeters?

(a) $1/10\pi$ cm/s (b) 2/5 cm/s

(c) 1 cm/s (d) 2 cm/s

225. The side of a square is increasing at a rate of 2 centimeters per second. How fast is the area of the square increasing when the side is 10 centimeters?

(a) 20 cm^2/s (b) 40 cm^2/s (c) 60 cm^2/s (d) 80 cm^2/s

226. A particle moves along a straight line with a velocity function $v(t) = t^2 - 3t$, where t is the time in seconds. When is the particle at rest?

(a) $t = 0$ and $t = 3$ (b) $t = 0$ and $t = 2$

(c) $t = 1$ and $t = 3$ (d) $t = 1$ and $t = 2$

227. The displacement of an object is given by $s(t) = 3t^2 - 4t + 2$. What is its acceleration at $t = 1$?

(a) 2 m/s^2 (b) 4 m/s^2 (c) 6 m/s^2 (d) 8 m/s^2

228. The displacement of a particle is given by $x(t) = 4t^3 + 3t^2 - 6t + 1$. What is its velocity at $t = 2$?

(a) 41 m/s (b) 3 m/s (c) 54 m/s (d) 21 m/s

229. If the position of an object is given by $x(t) = 2e^t + 5$, what is its velocity?

(a) $2e^t$ (b) $2e^t + 5$ (c) e^t (d) $e^t + 5$

$V = \frac{dx}{dt} = 2(e^t) + 0 = 2e^t$

223

$A = \pi r^2$

$\frac{dA}{dr} = \frac{d(\pi r^2)}{dr} = \pi 2r$

224

$V = \frac{4}{3}\pi r^3$

$\left(\frac{dv}{dt}\right) = 10\ cm^2/sec$

$\left(\frac{dr}{dt}\right) = ??$

$r = 5\ cm$

$\frac{dv}{dt} = \frac{4}{3}\pi 3r^2 \frac{dr}{dt}$

$\frac{dv}{dt} = 4\pi r^2 \frac{dr}{dt}$

$10 = 4\pi 25 \frac{dr}{dt}$

$\frac{dr}{dt} = \frac{1}{10\pi}$

$V = 4(3t^2) + 3(2t) - 6 + 0$

$V = 12(2)^2 + 6(2) - 6$

$= 48 + 6 = 54$

Maxima & minima :— of 'y' at different x

at maxm

" $\frac{dy}{dx} = 0$ "

$\frac{d^2y}{dx^2} = -ve$

at max



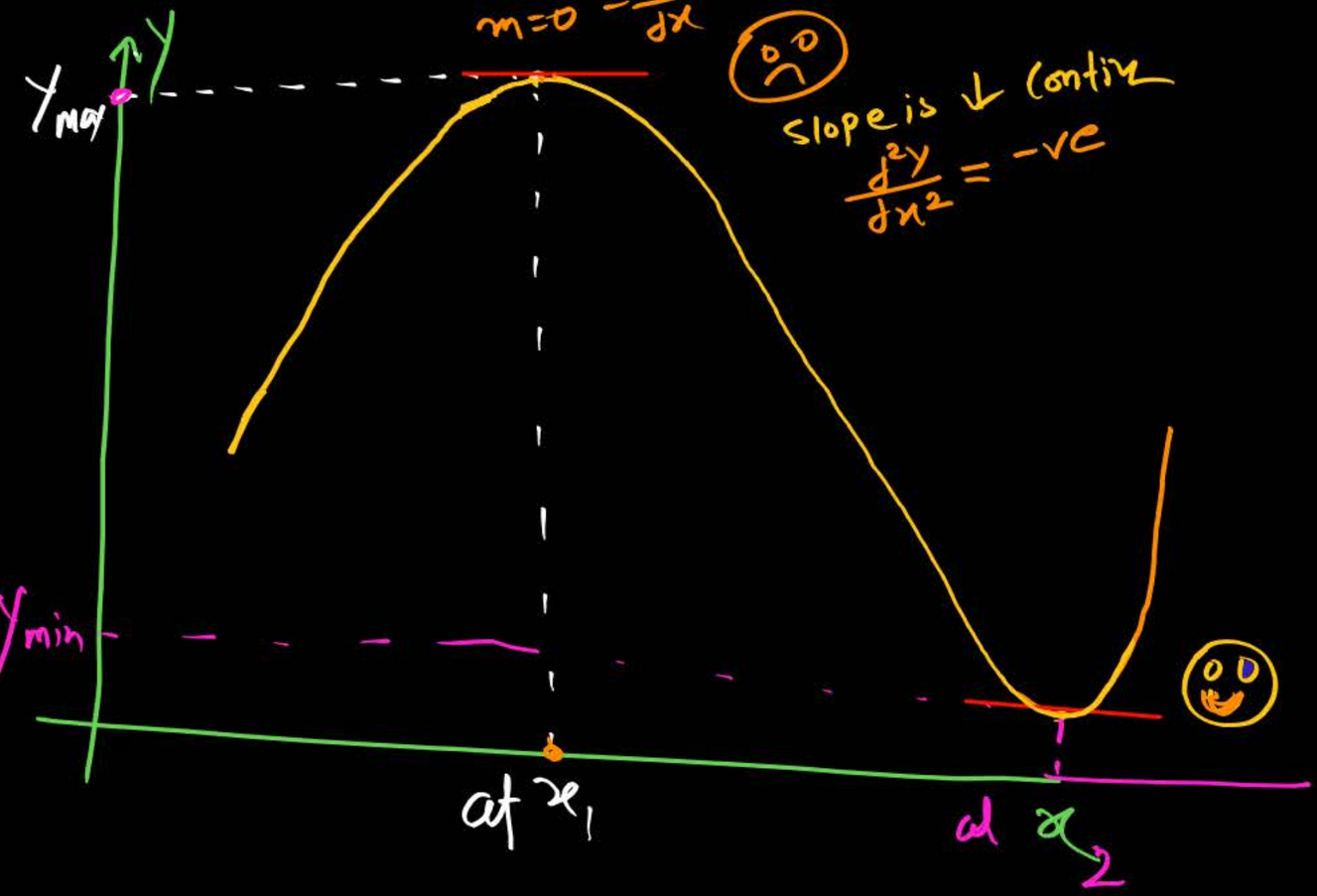
at minima

$\frac{dy}{dx}$ (slope) $= 0$

$\frac{d^2y}{dx^2} = +ve$



Change in slope $\left(\frac{d^2y}{dx^2}\right) = +ve$



MR* Box for maxima/minima

Step-1 Put $\left(\frac{dy}{dx}=0\right)$ & find

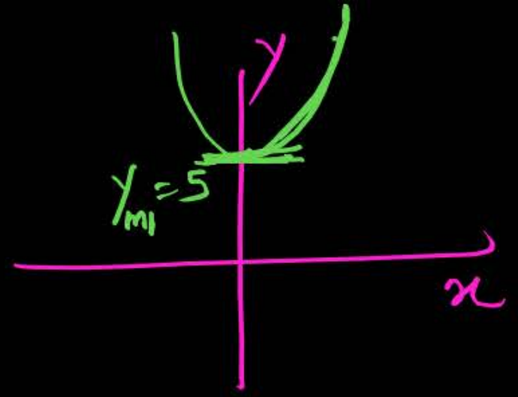
x where y will be max^m or min

Step-2 $\frac{d^2y}{dx^2}$ = +ve ← min^m ☺

$\frac{d^2x}{dx^2}$ = -ve ← Maxi^m ☹

Question

Find maxima and minima of $y = x^2 + 5$



$y = x^2 + 5$

→ for max^n/min of Y

$\frac{dy}{dx} = 0$

$\frac{dy}{dx} = 2x + 0 = 0$

$2x = 0$

$x = 0$ ✓

$\left(\frac{dy}{dx}\right) = 2x$

$\left(\frac{d^2y}{dx^2}\right) = +2$

minm at $(x=0)$

y will be minm at $(x=0)$

$y = x^2 + 5$

$y_{(at\ x=0)_{min}} = 0 + 5 = +5$

Question

Find maxima and minima of $y = 4x - x^2$

Step-1

[y will be max^m or min^m at $\frac{dy}{dx} = 0$]

$$\frac{dy}{dx} = 0 = 4 - 2x$$

$$0 = 4 - 2x$$

$$2x = 4$$

$$x = 2$$ → slope is zero at $x = 2$

$$\frac{d^2y}{dx^2} = 0 - 2 \times 1$$

$$\left|\frac{d^2y}{dx^2}\right| = -2$$

max

$$(y)_{\text{max at } x=2} = 4(2) - (2)^2$$

$$= 8 - 4 = 4$$

(Q) $y = x^3 - 3x^2 + 4$ find maximum and minimum value of 'y' and x at which y will be max^m.

Soln y will be max or min $\frac{dy}{dx} = 0$

$$\frac{dy}{dx} = 3x^2 - 3(2x) + 0 = 0$$

$$= 3x^2 - 6x = 0$$

$$= 3x(x-2) = 0$$

$x_1 = 0$ | $x_2 = 2$

$$\frac{d^2y}{dx^2} = 3(2x) - 6 = 6x - 6$$

$\left|\frac{d^2y}{dx^2}\right| = -6$ at $x_1 = 0$ → at $x_1 = 0$ Y will be maxim

$\left|\frac{d^2x}{dx^2}\right|_{x_1=2} = 6x - 6 = 6\times2 - 6 = 12 - 6 = +6$ ← min

$y_{max\,(at\,x=0)} = 0 - 0 + 4 = 4$ Ans

$y_{min\,(x_1=2)} = 2^3 - 3(2)^2 + 4 = 8 - 12 + 4 = 0$

MR* Y will be max or min at $\frac{dy}{dx} = 0$

$\frac{dy}{dx} = 3x^2 - 3(2x) = 0$

$3x^2 - 6x = 0$ | $x_1 = 0$, $x_2 = 2$

$3x(x-2) = 0$

(max) Y at $x_1 = 0 = +4$

(min) Y at $x_2 = 2 = 8 - 3(4) + 4 = 0$

Question

If velocity $V = t^3 - 6t^2 + 12$, then find maxima and minima value of velocity.

at $\frac{dV}{dt} = 0$ V will max or min

$$\frac{dV}{dt} = 0 = 3t^2 - 6(2t) + 0$$

$$0 = 3t^2 - 12t$$

$$0 = 3t(t-4)$$

$t_1 = 0$ $t_2 = 4$

$V_{t_1=0} = +12$ max

$V_{t=4} = 64 - 6 \times 16 + 12$

min $= 76 - 96 = -20$

Question

If $y = x^2 - 4x + 5$, then find maxima and minima of 'y'.

y will be max^n or min^n at $\frac{dy}{dx} = 0$

$2x - 4 + 0 = 0$

$2x - 4 = 0$

$\boxed{x = 2}$

$y_{min}\,(x=2) = 4 - 4 \times 2 + 5$

$= 9 - 8 = 1$

$$y = x^3 - 12x^2 + 36x$$

find maximum and minimum Value of "y"

$$\frac{dy}{dx} = 3x^2 - 12(2x) + 36 = 0$$

$$3x^2 - 24x + 36 = 0$$

$$3(x^2 - 8x + 12) = 0$$

$$x^2 - 8x + 12 = 0$$

$$x^2 - 6x - 2x + 12 = 0$$

$x_1 = 2$ $x_2 = 6$

$$y_{x_1=2} = 8 - 12(4) + 36 \times 2$$

$$= 8 - 48 + 72 = 80 - 48$$

$$= 32$$

$$y_{x_2=6} = 216 - 12(36) + 36 \times 6$$

$$\text{min} = 0$$

$y_{min} = 0$ at $x_2 = 6$

$y_{max} = 32$ at $x_1 = 2$



Thank

You